

1. (a) The curve $y = kx^2 + 3x + k - 4$, where k is a constant, lies completely above the x -axis.
Find the range of the values of k . [4]

- (b) Find the range of values of m for which the line $y + 6 = -mx$ is a tangent to the curve
 $y = x^2 + 3x + 2m$. [4]

2. The polynomial $f(x) = 4x^4 - 6x^3 - 4p^2x^2 + px - 7$, where p is a constant, is exactly divisible by $x - p$.

(a) Show that $6p^3 - p^2 + 7 = 0$. [2]

(b) Show that $p = -1$ is the only real solution of the equation in part (a). [5]

3. Solve the simultaneous equations

$$27^y \times 3^x = 1,$$

$$2^{y-2} \div 4^x = 32^{\frac{1}{y}}.$$

[5]